

apt & dpkg Cheat Sheet

Install, update and remove software on Debian and Ubuntu, and find out which package owns a file.

Using it

```
sudo apt update    ← refreshes the package lists only; installs nothing
sudo apt upgrade   ← installs newer versions; never removes a package
sudo apt full-upgrade ← may also remove or replace packages to finish (big updates)
```

Command	Does	Command	Does
apt install pkg	Installs it plus what it needs	apt search word	Searches all packages
apt install ./app.deb	Local file + its deps	apt show pkg	Version, size, depends
apt install pkg=1.2-3	A specific version	apt policy pkg	Installed vs available
apt remove pkg	Removes it, keeps config	apt list --installed	Everything installed
apt purge pkg	Removes it and its config	apt list --upgradable	What can be updated
apt autoremove	Removes unneeded deps	apt-mark hold pkg	Freezes its version
apt install --reinstall pkg	Puts back deleted files	apt-mark showhold	Lists held; unhold frees
dpkg -L pkg	Files the package installed	dpkg -S /usr/bin/dig	Owner: bind9-dnsutils

Reading apt policy nginx

```
nginx:
  Installed: 1.24.0-2ubuntu7 ← what you have now ((none) = not installed)
  Candidate: 1.24.0-2ubuntu7.1 ← what install/upgrade would give you
  Version table:
     1.24.0-2ubuntu7.1 500 ← version, then priority: highest wins (500 = normal repo)
        500 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages
 *** 1.24.0-2ubuntu7 500 ← *** = the installed one
        100 /var/lib/dpkg/status ← 100 = the copy on this machine
```

Reading dpkg -l

/	Name	Version	Architecture	Description
ii	nginx	1.24.0-2ubuntu7	amd64	small, powerful,... ← installed and OK
rc	apache2	2.4.58-1ubuntu8.4	amd64	Apache HTTP Server ← removed, config left: purge
iU	mytool	3.0-1	amd64	In-house tool ← unpacked, not set up: broken

1st letter = wanted (**i** install, **r** remove, **h** hold) · 2nd = actual (**i** installed, **c** config only, **U** unpacked) · uppercase = trouble

Fixing broken installs

Error says	Do
"dpkg was interrupted..."	sudo dpkg --configure -a
"Unmet dependencies. Try 'apt --fix-broken install'"	sudo apt --fix-broken install
"Could not get lock /var/lib/dpkg/lock-frontent"	Another apt is running (often auto-updates): wait. Never delete the lock

Where packages come from

Ubuntu 24.04+ / Debian 13+: `/etc/apt/sources.list.d/*.sources` in deb822 format (**URIs:**, **Suites:**, **Components:**, **Signed-By:** lines). Debian 12: one-line **deb URL bookworm main** in `/etc/apt/sources.list`.

Handy recipes

Command	Does
ls /var/run/reboot-required	Exists = reboot needed (new kernel or libc)
less /var/log/apt/history.log	What was installed or removed, and when
apt-cache rdepends --installed libssl3	Installed packages that depend on it
sudo dpkg-reconfigure -pnow unattended-upgrades	Auto security updates (Ubuntu: on by default)

Remember: **update** refreshes the list, **upgrade** installs; run **update** first. Check what will be removed before saying yes.